

Utilizing Everyday Textures as Concept-Based Explanations for Deep Learning Model Prediction of Glioma Biomarker Status

Rayyan Akhand¹, Farzad Alizadeh², Thanh Binh Nguyen³, Nick D. James⁴,
Sreeraman Rajan⁵, *Senior Member, IEEE*, and Rebecca E. Thornhill¹

Abstract—Objective: This study aims to bridge the gap between artificial intelligence (AI) model prediction and human comprehension of higher level imaging concepts. The specific application involves magnetic resonance imaging (MRI) of gliomas, focusing on AI classification of isocitrate dehydrogenase (IDH) biomarker status, a key indicator of glioma aggressiveness. **Methods:** Four convolutional neural network (CNN) models, based on the ResNet50 architecture, were trained using MRI of gliomas to differentiate between gliomas with “wildtype” and “mutated” forms of the IDH biomarker. The models were evaluated using fivefold cross-validation. In addition, the testing concept activation vectors (TCAV) framework was employed to assess the impact of “everyday” image texture concepts on model decision-making. Sensitivity scores ranging from 0 to 1 were calculated for 47 concepts across three layers of each model for both forms of IDH. **Results:** The analysis revealed varying levels of sensitivity to different image texture concepts when classifying IDH forms. These sensitivity scores indicate the extent to which the model is influenced by these concepts when making decisions about the IDH type. **Conclusion:** This study provides a preliminary evaluation of how concept-based explanations can elucidate the internal workings of closed-box neural networks, particularly in a medical context. **Significance:** The findings offer valuable insights into how concept-based explanations can improve understanding and trust in AI models, which is crucial for deploying AI models in clinical settings.

Impact Statement—Few aspects of healthcare technology remain untouched by AI, and radiology is no exception. Although the uptake of some AI applications has been uncontroversial,

there is justifiable reluctance to deploy complex models in critical diagnostic tasks. For example, deep networks trained on large imaging datasets can delineate between stable and aggressive forms of glioma. While encouraging, we still lack visibility into how deep models reach their decisions. In this study, we adapted the testing with concept activation vector (TCAV) method to better understand how simple image concepts called “describable textures” are represented by deep models trained on magnetic resonance images acquired from patients with gliomas. Our results reveal an inverse relationship in the TCAV scores associated with stable lesions and those associated with aggressive gliomas. With further concept refinement, this approach could be leveraged to enable communication of model decisions in medical imaging to referring physicians and their patients.

Index Terms—Concept-based explanations, deep learning, explainable artificial intelligence (AI), glioma, medical imaging.

I. INTRODUCTION

GLIOMAS are a rare but deadly form of cancer in the central nervous system (CNS). Conventional radiologic assessment of gliomas on medical images such as magnetic resonance (MR) images can vary based on individual experience and expertise. To address the variability in radiologic assessment, there is a growing interest in leveraging artificial intelligence (AI) to analyze medical images to classify glioma subtypes [1], [2].

Despite promising predictive performances, the clinical deployment of AI algorithms faces hurdles, chiefly due to their real and perceived closed-box nature. Understanding what drives AI model decisions is crucial for establishing trust in AI-based diagnostics. Although several approaches have been explored to explain the decision-making mechanisms of AI models, the vast majority emphasize describing predictions based on input features. These features, such as individual pixel values in input images, are often considered to be “low level” and are often ill-suited for capturing the broader context of human perception. Research in psychology and neuroscience has long demonstrated that human cognition is based on perceptual symbols that extend beyond sensory input to capture higher level conceptual knowledge [3]. As such, an ideal framework for explainable AI (XAI) would align with humans’ natural ability to grasp high-level abstractions. Textural properties are high-level concepts that offer a rich source of visual information, especially for describing objects that display patterns [4].

Received 16 April 2025; revised 25 February 2026; accepted 1 April 2026. Date of publication 7 April 2026; date of current version 31 August 2026. This work was supported in part by the Natural Sciences and Engineering Research Council of Canada (NSERC), under Grant RGPIN-2021-04224. The work of Rayyan Akhand was supported in part by an Ontario Graduate Scholarship. This article was recommended for publication by Associate Editor Alejandro F. Frangi upon evaluation of the reviewers’ comments. (*Corresponding author: Rebecca E. Thornhill.*)

Rayyan Akhand was with the Department of Systems and Computer Engineering, Carleton University, Ottawa, ON K1S 5B6, Canada. He is now with Schulich School of Medicine and Dentistry, Western University, London, ON N6A 5C, Canada (e-mail: rakhand@uwo.ca).

Farzad Alizadeh and Sreeraman Rajan are with the Department of Systems and Computer Engineering, Carleton University, Ottawa, ON K1S 5B6, Canada (e-mail: farzadalizadeh@email.carleton.ca; sreeramnr@sce.carleton.ca).

Thanh Binh Nguyen and Rebecca E. Thornhill are with the Department of Radiology, Radiation Oncology, and Medical Physics, University of Ottawa, Ottawa, ON K1Y 4E9, Canada (e-mail: thnguyen@toh.ca; rthornhill@toh.ca).

Nick D. James is with the Department of Analytics, The Ottawa Hospital, Ottawa, ON K1Y 4E9, Canada (e-mail: njames@toh.ca).

Digital Object Identifier 10.1109/TAI.2026.3681537

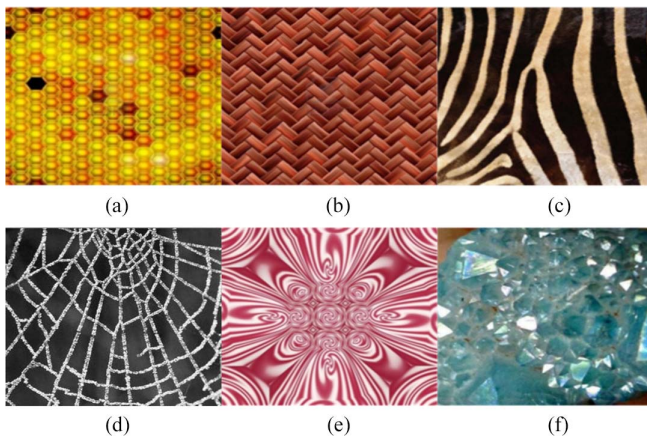


Fig. 1. Sample images of different texture concepts seen in day-to-day life. (a) Honeycombed. (b) Woven. (c) Striped. (d) Cobwebbed. (e) Swirly. (f) Crystalline. Images are taken from [4].

In this study, everyday textures are proposed as foundational image “concepts” and are used to reveal the decision-making process of a convolutional neural network (CNN) through “testing with concept activation vectors” (TCAV) [5]. This method examines the impact of each concept on the internal representations of the CNN, commonly utilized in medical AI research. Here, the term “concept” is defined by the set of representations corresponding to a specific feature of interest. Fig. 1 showcases images exemplifying six image texture concepts that are frequently observed in daily life (i.e., “everyday textures”) [4]. By quantifying the degree of activation induced in specific layers of a neural network by these everyday textures, one can provide insights into the key concepts driving model predictions.

This study aims to utilize everyday textures as concept-based explanations for deep learning model prediction of glioma biomarker status. Due to the significance of the isocitrate dehydrogenase (IDH) enzyme on patient prognosis, the article focuses on explaining models trained to delineate gliomas with “wildtype” forms from those exhibiting a mutant form of IDH biomarker. Patients with the mutated form of IDH are more likely to experience longer survival and better outcomes than those with the wildtype form [6], [7].

In this article, the gaps between AI model explanations and human understanding of high-level concepts are addressed through three key contributions. As the first contribution, magnetic resonance imaging (MRI) sequence-specific deep learning models designed to differentiate between gliomas with IDH wildtype (IDH-wt) and IDH mutant (IDH-mut) biomarker status are developed. A framework for evaluating concept activation vector (CAV) scores, using everyday texture image concepts to explore fundamental patterns that influence the decision-making process of each model is proposed and this forms the second contribution. In the third contribution, the differences in concept activation scores between the IDH groups across four types of MRI contrasts: apparent diffusion coefficient (ADC) Maps, T1-weighted postcontrast (T1c), T2-weighted (T2), and fluid-attenuated inversion recovery (FLAIR)

are investigated. These contributions aim to offer insights into the underlying MR image features that guide AI-based glioma classification.

II. LITERATURE REVIEW

AI has attracted a great deal of interest in the field of neuro-oncology with a rapidly increasing number of publications regarding deep neural networks (DNNs) on glioma imaging in the past two decades [2]. One of the strongest concerns regarding DNNs relate to difficulties inherent to their explainability or interpretability. The large number of model parameters and multilayered nonlinear mappings from input space cause these models to become incredibly complex and difficult to explain and interpret [2]. While explainability pertains to understanding the reasoning behind a model’s decisions, interpretability delves deeper into elucidating the specific intricacies of how a model operates. Explainability is often prioritized for practical implementation and deployment, as it plays a crucial role in ensuring that end-users understand and trust the decisions made by the model [8].

While various types of explanations exist, radiologists (physicians with specialty training in diagnostic imaging) are particularly interested in understanding image features that align with established medical ontologies [2], [9]. Feature-based explanations, such as saliency maps, are valuable for visualizing the important features within the input image that contribute to model predictions. For instance, Ahmad et al. [10] utilized deep learning networks to delineate which regions of a given input T2 weighted MRI image are the most discriminative and created class activation maps (CAMs). The CAM method is a popular XAI method used in medical imaging applications and employs a global average pooling (GAP) layer at the end of a CNN model from which the weighted sum of internal activation maps is calculated for a given layer and used to generate a saliency map of the same dimensions [11]. By creating these saliency maps, the CAM method assigns relative importance to each individual pixel for a given image. These CAMs were presented as color maps superimposed onto the original images and conveyed how tumoral regions were important for classification of IDH-mut while peri-tumoral edema regions were also significant for IDH-wt classification.

Gradient-based methods provide another approach to explain deep learning model predictions using feature attributes [2]. Gradient-weighted class activation map (Grad-CAM) build on the CAM technique by backpropagating gradients from the predicted class score through the feature maps in the final convolutional layer [12]. The Grad-CAM approach is advantageous over CAMs as it has shown to confer more precise localization of relevant image regions on saliency maps by leveraging gradient information [12], [13]. By utilizing Grad-CAM on four different deep learning models, Mercaldo et al. [14] demonstrated that the method localizes the areas within the tumor that contribute to the model’s prediction. A subsequent subjective evaluation by expert radiologists confirmed that the highlighted regions corresponded to the correct pathological areas.

Layerwise relevance propagation (LRP) is a technique that provides visual explanations for decisions made by DNNs, differing from CAM methods in both approach and applicable architectures. Following the prediction of the class of an input image, LRP computes the relevance score for each input feature based on its contribution to the final prediction score using its weights and the activation function. This process involves back-propagating the relevance score from the output layer through the layers of the network to identify influential pixels. The versatility of the LRP method is attractive, in that it can be applied to various types of neural networks, whereas CAM is tailored specifically for CNNs with a GAP layer followed by a fully connected layer. The LRP method has proven effective in explaining the decisions made by DL algorithms in glioma classification studies [15]. Sefcik and Benesova [16] proposed utilizing LRP not only to visualize model decisions, but also to improve the model using “explanation-guided training.”

Prototype-based methods, such as Prototype networks offer an intuitive approach to XAI, mimicking human cognition with a “this looks like that” method [17]. These methods classify data by comparing new data points to prototypes from the training set using similarity measures. Part-prototype networks focus on specific image parts, enhancing context and intuitiveness in how models learn. These methods resemble human analogical reasoning when explaining previous prototypes [18]. Chang et al. [19] applied this approach to glioma classification, using prototypical images identified through principal component analysis. However, prototype-based networks can show discrepancies between the prototype and the activation region of the query, leading to explanations that may lack interpretability [20].

Feature-based XAI methods often struggle to provide clear global explanations, as they can become too focused on low-level features. In contrast, concept-based approaches, such as those utilizing CAVs [5], leverage high-level concepts to enhance interpretability. This method allows new user-provided data to be introduced as a “concept” related to the input data of a given DL model. The CAV is orthogonal to the decision boundary established by a linear classifier that distinguishes between concept examples and random counterexamples for the activations at a specified, intermediate layer. Hence, computing the directional derivative of the network logit with respect to the activation at that layer, along the direction of the CAV reveals the relative importance of the concept in the model’s prediction sensitivity. As a result, TCAV allows for: customization, by allowing users to define desired concepts for measurement; seamless integration, by facilitating explanations without having to modify or re-train models; and globality, by ensuring a single global attribution score that may be used to interpret entire classes or sets of examples.

This study, therefore, proposes the implementation of TCAV to provide customizable, global explanations for a clinical prediction task. By utilizing a wide range of everyday textures as image concepts, the importance of each concept for CNN classification of IDH-wt and IDH-mut gliomas will be evaluated. By training a DL-CNN model on various MRI sequences taken from a public glioma dataset, texture concepts will be

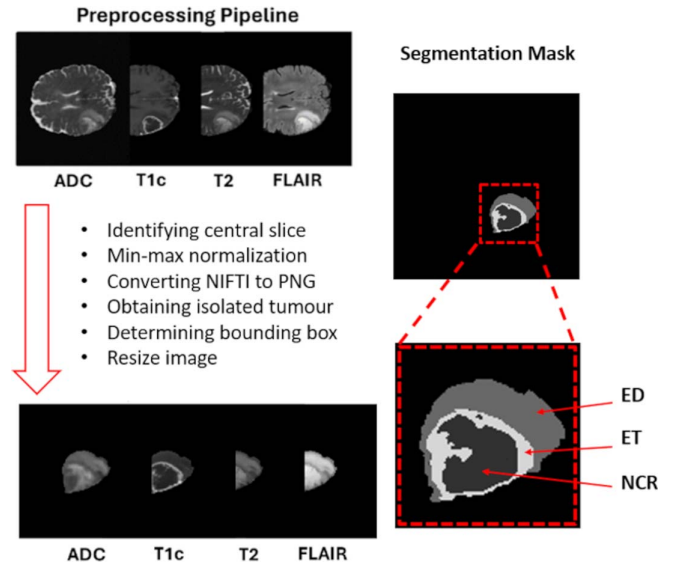


Fig. 2. Summary of preprocessing steps performed on the UCSF-PDGM dataset. NCR, nonenhancing and necrotic tumor core; ET, enhancing tumor; ED, edematous tissue.

scored based on the activation vector produced within early, intermediate, and late layers of each CNN.

III. METHODOLOGY

A. MRI Dataset

Four hundred and ninety-six subjects were selected from the University of California San Francisco Preoperative Diffuse Glioma MRI dataset (UCSF-PDGM), a publicly available labelled dataset of MRI exams, which includes MRI sequences typical of a preoperative MRI protocol [21]. Four of these sequences namely ADC, T1c, T2, and FLAIR were selected for each subject. The dataset accounted for bias correction, artifact removal, and registration to ensure proper spatial alignment of MRI slices between and among sequences obtained for each subject. Moreover, the image dataset is accompanied by tumour segmentation masks, which include delineation of all three major tumor compartments: a central nonenhancing, and necrotic tumor core (NCR), surrounded by an enhancing tumor region (ET), which was in turn surrounded by an edematous region (ED, denoted by the extent of the abnormality on FLAIR images).

B. Image Preprocessing

Fig. 2 portrays the entire preprocessing pipeline with a representative case from the UCSF-PDGM dataset. To prepare the data for input into a 2-D ResNet50 model, a single, central slice was selected for each subject from each 3-D stack of ADC, T1c, T2, and FLAIR sequences. The criterion for central slice selection was defined as the slice with the largest NCR area. This criterion was implemented due to the remarkable heterogeneity found in the NCR, which may provide a richer source of patterns for the DL model to learn from. To accomplish this, the number of pixels corresponding to the NCR across all slices within the tumour segmentation mask was calculated, and the

slice with the most NCR pixels was chosen. For instances where an NCR was absent, which was true for 35 cases, the largest overall tumor area was used instead. One patient in the IDH-wt group was missing a segmentation mask and was therefore excluded from the study to yield a final count of 391 IDH-wt and 103 IDH-mut exams (a total of 494).

Each selected central slice was normalized using min–max scaling. The normalized 2-D slice was subsequently converted from NIfTI to portable network graphics (PNG) format for downstream processing. For each central slice, any pixel contained within the full extent of the tumor (ED, ET, or NCR) was included in the final, isolated tumor region, and pixels without (i.e., surrounding brain tissue) were excluded. The boundaries of these regions were determined by counting the number of pixels between the first and last nonzero-pixel instance along each row (x-direction, L_x) and column (y-direction, L_y). The largest dimension identified from the entire collection of tumors, either L_x or L_y , was designated as $L_{\max}$. This dimension was then used to create a square bounding box ($L_{\max} \times L_{\max}$) which was subsequently applied to all 1976 images (494 subjects $\times$ 4 MRI sequences), ensuring that each segmented tumor was positioned as close to the center of this box as possible. Each grayscale tumor image was then resized to 224×224 pixels. To satisfy the input requirements of the ResNet50 architecture, the single-channel grayscale image was duplicated across three channels to produce a $224 \times 224 \times 3$ input tensor.

C. Development of DL Models for the Prediction of Glioma IDH Status

All ResNet50-based models described were developed and trained using Python, employing the Keras package with a Tensorflow backend. Models based on the ResNet architecture were chosen as they exhibited resilience in deeper layers, addressing the challenge of vanishing gradients [22], [23], and extensively applied in medical imaging classification (including IDH classification) [10], [15], [24], [25], [26], [27], [28], [29]. Four distinct 2-D ResNet50 models were generated, each tailored to analyze a specific MRI contrast (ADC, T1c, T2, and FLAIR), producing softmax probabilities indicative of the IDH-wt or IDH-mut class membership. The positive class corresponded to the IDH-mut group.

The “Hyperband” algorithm was utilized to optimize key hyperparameters, including the number of units in the densely connected layer, batch size, and dropout rate for each of the sequence-specific ResNet-based models [30]. The entire dataset was split into a train and hold-out test set, with a test set size of 5% ($n = 24$) due to the limited available data. The train set was then further partitioned into training and validation sets for each fold of stratified cross-validation to refine the hyperparameter search. For each model, all slices and images belonging to a subject were assigned exclusively to a single partition (train, validation, and test). No subject appeared in more than one partition.

D. Evaluation of DL Model Performance

After selecting the optimal hyperparameters, the final model for each MRI sequence was trained on the entire training set

and evaluated on the independent hold-out test set (5% of the dataset, $n = 24$). A decision threshold of 0.5 was used for all models. Model performance on the hold-out set was assessed using accuracy, precision, recall, and F1-score. Precision, recall, and F1-score were computed using macro-averaging. In each case, 95% confidence intervals were estimated via the percentile interval bootstrap method [31] using 1000 repetitions to account for instance-level sampling variability arising from the small test set. The confidence intervals are conditional on the trained model and selected hyperparameters. They do not capture variability arising from model training stochasticity or hyperparameter selection (i.e., training or selection variability). The development, training, and evaluation stages are depicted in Algorithm 1 (Appendix).

E. Evaluation of Concept Activation Vector Scores Using Everyday Texture Image Concepts

The steps involved in the evaluation of TCAV scores are depicted in Algorithm 2 (Appendix). To generate and test concept activation vectors, both concept-representing images and random negative counterexamples are required. For this study, concept images from the Describable Textures Dataset (DTD) [4] were used. This dataset contains 47 categories of “human-friendly” texture attributes, providing a diverse range of everyday patterns. Each category is annotated with descriptive adjectives such as “cobwebbed,” “freckled,” and “striped.” Negative counterexamples were obtained from the Microsoft Common Objects in Context (MS-COCO) dataset [32], consisting of 5000 images. In this context, “random” simply refers to the absence of a suitable common descriptor that would appropriately characterize the entire set of images.

To generate activations for both concepts and random examples, the 5640 DTD and 5000 random images underwent resizing to meet the input dimensions required for the ResNet50 models ($224 \times 224 \times 3$). The workflow for implementing the TCAV methodology was implemented with the assistance of code published by Kim et al. [5], which was modified for use in the current study, and includes the following steps.

- 1) *Instantiating a Model Wrapper*: A Model Wrapper class was created to connect our model to the `tcav.py` file, which extracts the internal information for generating activations. Inside this class, the method `get_bottleneck_tensors` retrieves specific internal tensors defined by the user and stores them in a dictionary. In this study, three bottlenecks were selected for activation analysis, which we refer to as the early, intermediate, and late layers.
- 2) *Creating an Activation Generator*: An Activation Generator class was created to load example data for specific concepts and to call the model wrapper for bottleneck tensors. This class computes CAVs by returning activations for both concepts and random images, creating a dictionary that links concepts, bottleneck layers, and their corresponding activations.
- 3) *Combining the Information*: Logistic regression classifiers are trained by using the activations from the previous

step to linearly separate the concepts, ultimately obtaining the CAVs. The TCAV process in a separate tcav.py file is then executed for a target class and a set of concepts. This step retrieves the CAV-directional derivative of the network logit with respect to the corresponding layer, along with the number of random experiments for assessing statistical significance. The result is a TCAV score for each concept and for each layer.

Mathematically, a CAV ($\mathbf{v}_l^C$) is just the parameter vector of a logistic regression model trained to distinguish between activations at layer l , which were generated by the concept of interest (C), from those generated by the random set [5]. Conceptual sensitivity ($S_{C,k,l}$) is the dot product between this CAV and the gradient of the network logit function ($\nabla \mathbf{h}_{l,k}(f_l(\mathbf{x}))$) taken with respect to the activation at layer l [6]

$$S_{C,k,l}(\mathbf{x}) = (\nabla \mathbf{h}_{l,k}(f_l(\mathbf{x}))) \cdot (\mathbf{v}_l^C / \|\mathbf{v}_l^C\|). \quad (1)$$

Here, $f_l(\mathbf{x})$ is defined as the activations for input $\mathbf{x}$ at layer l , and $\mathbf{h}_{l,k}$ maps the activation vector to the network's logit for class k . A positive conceptual sensitivity is the result of the gradient and CAV pointing in the same direction, implying that an increase in the activation of concept C leads to an increase in the network's logit values for class k . In other words, when there is an increase in the presence of concept C in the input data $\mathbf{x}$, there is higher likelihood or confidence of predicting class k [33].

For instance, consider the IDH-wt class, where data labeled as IDH-wt are collected, and conceptual sensitivity is calculated for each input image. The TCAV score for a specific concept, such as “bubbly,” reflects the ratio of IDH-wt images with positive conceptual sensitivities to the total number of wildtype images

$$\text{TCAV}_{\text{bubbly}, \text{wt}, l} = \frac{|\{\mathbf{x} \in X_{\text{wt}} : S_{\text{bubbly}, \text{wt}, l}(\mathbf{x}) > 0\}|}{|X_{\text{wt}}|}. \quad (2)$$

The Welch's t-test was used to assess differences in mean TCAV scores between the IDH-wt and IDH-mut groups for each concept, assuming a normal distribution with unequal variances. To minimize Bonferroni correction was applied to control the false discovery rate [35], with the adjusted α calculated as $\alpha_{\text{adjusted}} = 0.05/47 = 0.00106$.

IV. RESULTS AND DISCUSSION

Classification accuracy, precision, recall, and F1-score are presented in Table I. Whereas the primary focus of this study is on improving the explainability of CNN-based models rather than developing a new, state-of-the-art classifier, a ResNet-50 architecture was employed as a well-established and widely used backbone for IDH status classification from MRI. The models demonstrate performance with the expected range reported by others with respect to the classification of IDH biomarker status of gliomas on MRI [24], [26], [27], [36], [37], [38], making these models suitable and realistic platforms for interpretability analysis.

Figs. 3 and 4 feature heatmaps contrasting DTD concept TCAV scores between different class types for each model. The

TABLE I
KEY PERFORMANCE METRICS FOR THE BEST MODELS TRAINED TO IDENTIFY IDH STATUS FOR THE TEST SET

Model	Metric	Score	95% Confidence Interval
ADC Model	Accuracy	0.882	0.720–1.000
	Precision	0.667	0.500–1.000
	Recall	0.8	0.725–0.925
	F1-Score	0.727	0.625–0.850
	AUC	0.875	
T1c Model	Accuracy	0.961	0.880–1.000
	Precision	1	1.000–1.000
	Recall	0.8	0.750–0.950
	F1-Score	0.889	0.800–1.000
	AUC	0.984	
T2 Model	Accuracy	0.882	0.739–0.900
	Precision	0.75	0.690–0.830
	Recall	0.6	0.500–0.750
	F1-Score	0.667	0.500–0.775
	AUC	0.825	
FLAIR Model	Accuracy	0.882	0.720–1.000
	Precision	0.667	0.500–1.000
	Recall	0.8	0.725–0.925
	F1-Score	0.727	0.625–0.850
	AUC	0.875	

asterisks next to TCAV scores denote values that are statistically significant from random inputs. The results indicate an inverse relationship between the TCAV scores for IDH-wt and IDH-mut predictions. Specifically, concepts that elicit high activation scores in certain bottleneck layers for one target class tend to produce low scores for the alternative class, and vice versa. For example, in ADC ResNet50 model [Fig. 3(a)], the TCAV score for the “honeycombed” concept in the early layer is 0.11 for predicting glioma IDH-wt, while its score for predicting IDH-mut is 0.83. This suggests that the model is sensitive to the “honeycombed” concept in the prediction of both classes. As expected, even for concepts relevant to the prediction task, the magnitude of this inverse relationship was not exact. For instance, a TCAV score of 0.3 for one class does not necessarily imply a TCAV score of 0.7 for the other.

Examining the TCAV scores produced by each of the four models for IDH-wt and IDH-mut classes across layers, beginning with the ADC-ResNet50 model for the IDH-wt target class [Fig. 3(a), red heatmap] consistently demonstrated a pattern in the middle bottleneck layer (conv3_block4_out). At this layer, multiple concepts exhibit TCAV scores that were statistically significantly higher than those produced by random inputs. Notably, the “woven” concept was one of two concepts which reliably exhibited high scores throughout the model, yielding conceptual sensitivities greater than 0.850 for early (0.859 ± 0.14 , p -value < 0.05), intermediate (0.909 ± 0.14 , p -value < 0.01), and late (0.974 ± 0.05 , p -value < 0.001) layers, suggesting that this concept was especially influential in IDH-wt prediction throughout the ADC model. Conversely, for the IDH-mut prediction, there is a clear inverse relationship where scores for the “woven” concept, which were markedly high in wildtype prediction, did not exceed 0.3 in any of the three bottleneck layers [Fig. 3(a), blue heatmap].

Transitioning to the T1c-ResNet50 model, for the IDH-wt, the “potholed” and “porous” concepts produced significant

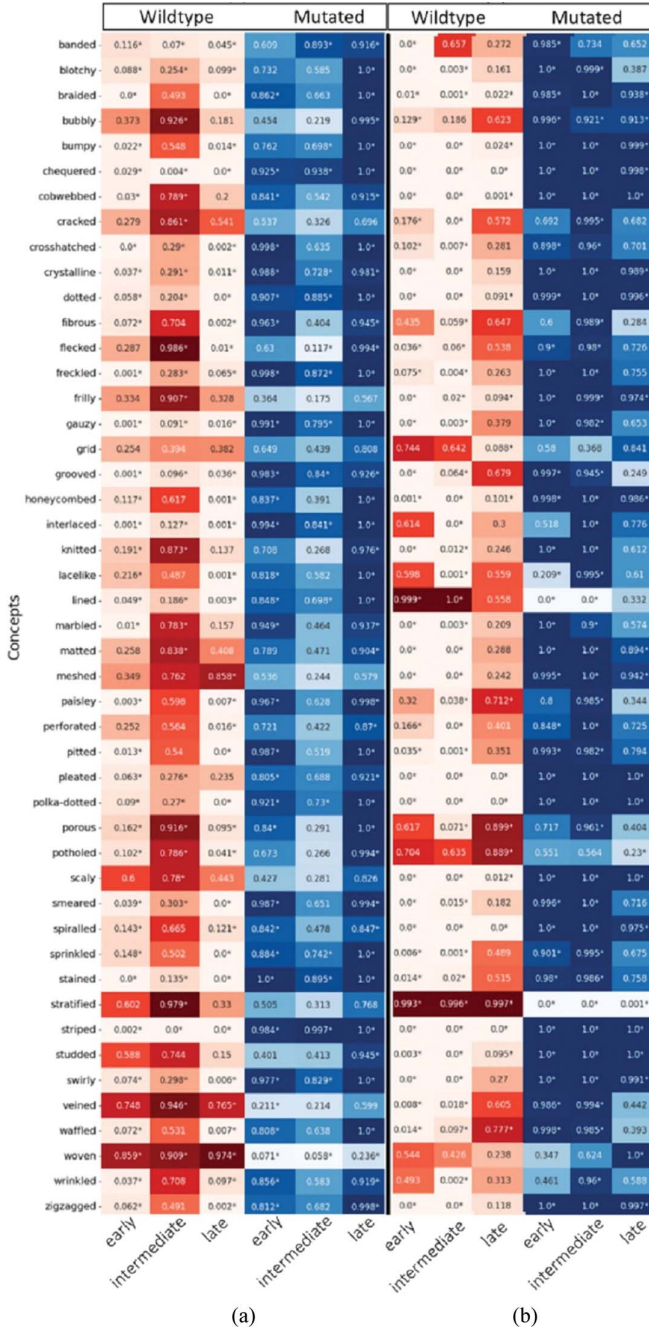


Fig. 3. Heatmap of the TCAV scores generated for IDH-wt (red) and IDH-mut (blue) classification by (a) ADC and (b) T1c ResNet50 models, for three of each model's bottleneck layers: conv2_block2_out (early), conv3_block4_out (intermediate), and conv5_block1_out (late). Cells with asterisks indicate concept scores significantly different from those generated by a set of random images.

TCAV scores ≥ 0.89 in late layer [Fig. 3(b)]. These findings are consistent with the visual descriptions of aggressive wildtype tumors which display heterogeneous perfusion and permeability [39]. The “stratified” concept displayed consistently high TCAV scores across all three layers, surpassing 0.99 throughout. This suggests the potential relevance of the “stratified” concept in the T1c model for IDH-wt classification. For the IDH-mut prediction [Fig. 3(b), blue heatmap] was

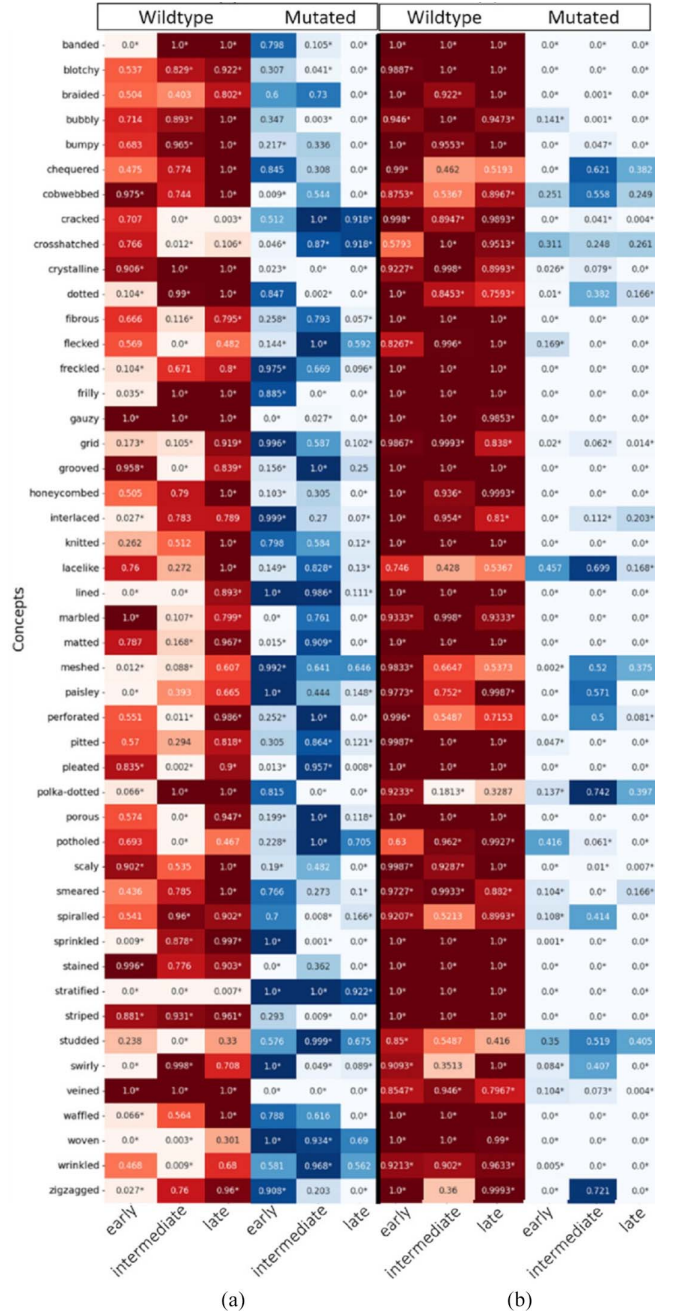


Fig. 4. Heatmap of the TCAV scores generated for IDH-wt (red) and IDH-mut (blue) classification by (a) T2 and (b) FLAIR ResNet50 models, for three of each model's bottleneck layers: early, intermediate, and late. Cells with asterisks indicate concept scores significantly different from those generated by a set of random images.

more densely populated with higher TCAV scores. Once again, concepts yielding standout high scores from the wildtype plot were notably lower than the rest of the concepts. For instance, the “stratified” concept produced TCAV scores of 0.00 ± 0.00 (p -value < 0.01) for all layers.

Compared to the stark differences in class distributions observed in the ADC and T1c TCAV scores, the distinction between IDH-wt and IDH-mut TCAV scores for the T2 model [Fig. 4(a)] was much more subtle. Several noteworthy concepts emerged, with three consistently achieving conceptual

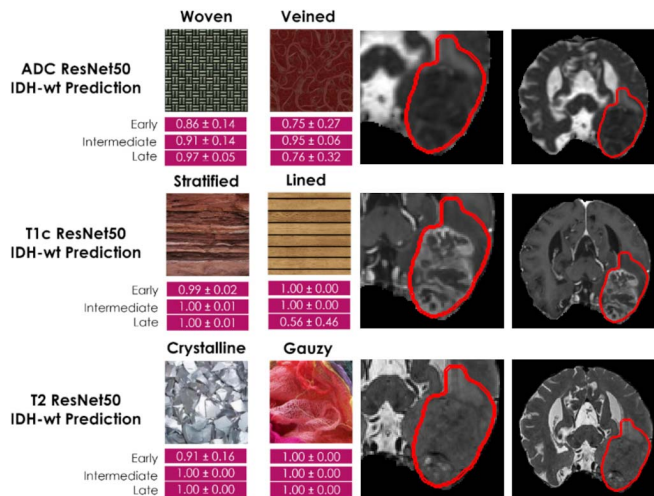


Fig. 5. Key concepts that exhibited high TCAV scores compared to other concepts within ADC, T1c, and T2 models for IDH-wt prediction.

sensitivities of 0.90 or higher across the model for IDH-wt prediction [Fig. 4(a), red heatmap]. These included the “crystalline,” “gauzy,” and “veined” concepts. The persistence of such high scores indicates a strong sensitivity of the model to these concepts when predicting the wild-type target. For IDH-mut classification, the “stratified” concept was the only one to yield significant TCAV scores of ≥ 0.90 across all three bottleneck layers. This finding is particularly interesting as it mirrors the significance of the “stratified” concept in the T1c model for IDH-wt prediction, yet here it is significant for the opposite, IDH-mut prediction.

The TCAV results associated with the FLAIR model’s wild-type prediction [Fig. 4(b), red heatmap] included TCAV scores ≥ 0.90 for nearly every single concept across all three bottleneck layers. The sole exception was the “polka-dotted” concept in the intermediate layer, indicating the insensitivity of the FLAIR model to this concept for IDH-wt prediction. By contrast, the TCAV scores associated with the IDH-mut predictions [Fig. 4(b), blue heatmap], rarely showed scores exceeding random (0.5), and those that did were statistically insignificant. These consistently low TCAV scores may indicate that the FLAIR model lacks sensitivity to the proposed concepts when making IDH-mut predictions.

Upon reviewing the results depicted in Figs. 3 and 4, a discernible pattern becomes apparent. Layers with higher proportion of significant concepts tend to display a greater proportion of significance between the mean TCAV scores of the two prediction types. For instance, in Fig. 4(b), the FLAIR model illustrates dense IDH-wt and sparsely distributed IDH-mut TCAV heatmaps, indicating a notably higher occurrence of significant values compared to nonsignificant ones, particularly evident within the early and late layers. This observation aligns with the considerably high percentage of concepts exhibiting differences in mean TCAV scores between IDH-wt and IDH-mut groups in the FLAIR model, particularly within the early (0.915) and late (0.851) layers. Similarly, the late layer of the T1c model displays a prevalence of concepts yielding insignificant TCAV scores [Fig. 3(b)]. These findings further reinforce

the earlier observation of an inverse relationship between IDH-wt and IDH-mut TCAV in terms of conceptual sensitivities. To better visualize distinct concepts, Fig. 5 presents examples of key concepts that exhibited high TCAV scores compared to other concepts for the ADC, T1c, and T2 models, and for predicting IDH-wt status.

V. CONCLUSION AND FUTURE WORK

This exploration of concept-based explanations has provided unique insights into AI decision-making in a relevant classification task in neuroradiology. By probing pretrained models with customizable “concepts,” the sensitivity of a deep model to everyday image concepts can be quantified using a single global metric, thus enhancing explanations beyond pixel-level interpretations. This approach has limitations, however, that should be addressed by future work.

We acknowledge the potential limitations of the CNN models generated for use in TCAV analysis. First, it is possible that training our models on the central slice of each lesion instead of on the entire 3-D tumor volume inflated the classification performance. Selecting the slice with the maximum NCR preferentially samples the most bulky or heterogeneous tumor areas. In clinical practice, visual detection of a tumor necrotic area on the T1-weighted images post contrast is key to the radiological diagnosis of a glioblastoma, which is the most frequent and aggressive form of IDH-wild type gliomas. Our rationale for the single slice selection strategy was that it provides a more direct comparison between what the radiologist sees and the imaging pattern which emerges from the deep learning model. In the literature, 2-D CNN models using 1 to 5 representative slices at or near the maximal tumor area have been reported to yield high accuracy in the classification of IDH-mutant vs wild-type gliomas [26], [40]. This suggests that the models created using the slices with maximum NCR were reasonable and representative platforms for interpretability analysis. Secondly, we recognize that the small size of the hold-out test sets for our models represents a potentially important limitation. Given that the dataset was very small from a deep learning perspective, we assessed that we needed to reserve as many cases as possible for training to produce a final model that would provide good classification performance.

The primary focus of our study is on improving the explainability of existing 2-D CNN-based models rather than developing a new state-of-the-art classifier. A key limitation of the TCAV approach is the assumption that activations corresponding to a concept are linearly separable from those of nonconcept (random) examples in the activation space. However, this assumption may not hold, particularly for complex or abstract concepts [41], [42], [43]. Deep learning models inherently capture intricate patterns through nonlinear relationships, enabling them to learn distributed representations. Information about complex concepts is spread across multiple neurons and layers, rather than being localized to a specific subset. This distributed nature challenges the assumption that a concept can be represented as a linear function within a single layer, as

TCAV implies. While lower level concepts (e.g., edges and textures) may be linearly separable, more abstract concepts require nonlinearity to capture their complexities and dependencies. One possible solution is the “Concept Gradients” framework proposed by Bai et al. [41], which generalizes from the fact that a CAV is simply the gradient of a *logistic regression* classifier’s logit at the point $f_l(\mathbf{x})$ (or at any other point), and that it can be seamlessly replaced by the gradient of any arbitrary differentiable classifier, most notably a CNN, which can respond to the spatial information in convolution layer activations that is otherwise lost when they are flattened to accommodate a linear classifier’s input requirements.

It is also important to stipulate that the TCAV score reflects only the correlation between the concept and the model’s prediction, not causality or any specific relationship, [44]. TCAV calculates the importance of a concept by taking the inner product of a CAV and the model’s input gradient. While TCAV scores can sometimes be interpreted [41] causally, other confounding factors may affect the results. Recognizing this distinction is crucial to avoid misleading conclusions about how concepts contribute to model decision-making.

One factor that limits conclusions regarding causality in this study is the redundancy among concepts derived from the DTD image set. Many concepts, such as “polka-dotted” and “dotted,” exhibited significant overlap in their descriptive properties, leading to similar activations and TCAV scores across layers. This overlap is understandable, given their perceptual similarity. The concept of “coherency” [45] highlights the importance of selecting concepts that are perceptually similar within a given concept but distinct from others. Future work could refine the set of image concepts, either with the aid of clustering techniques [45], or via surveys of end-users, or via a combination of both quantitative and qualitative methods.

Second, while TCAV assumes independence among tested concepts, real-world concepts often exhibit varying degrees of correlation. This interdependence makes it harder to discern the true importance of concepts for model predictions. A deeper challenge lies in concept selection for TCAV, as it relies on user-defined queries for the concepts of interest. While customizability offers advantages over other approaches, it also poses challenges, demanding hand-labelled examples and introducing bias. Relevant concepts can be overlooked, resulting in explanations that are incomplete.

Indeed, the current study suggests that presenting TCAV scores in isolation or out of context may result in misleading explanations for deep models. As shown in Section IV, many concepts exhibit high scores across a model for a given prediction type. Despite high scores, this uniformity does not imply equal contribution to the model’s prediction but rather alignment with internal activations. Moreover, these scores may not be sufficient to completely explain the model, as there is a possibility that concepts vital to the prediction have not yet been identified. Thus, future concept refinement should be informed by an assessment of completeness, for example via evaluation of a “completeness score,” which expresses the strength of the relationship between the set of chosen concepts and model prediction [45].

Ultimately, the overarching goal of this study is to ensure that end users of these AI models, including physicians and other healthcare professionals, can effectively interpret the findings derived from concept-based explanations. Future investigations should aim to determine whether clinicians believe concept-based explanations enhance their understanding of DL model predictions, gather feedback on the usefulness of the presented concepts, and incorporate clinical recommendations on relevant concepts for classification tasks.

In summary, the results from this study show promise in explaining DNNs in terms of human friendly concepts. With the burgeoning novelty of this field and myriad uncharted pathways, it is anticipated that concept-based explanations will shed light on the intricate workings of deep models and enable their potential deployment in clinical settings.

APPENDIX

Algorithm 1: Train IDH Classifiers.

Require: IDH Dataset preprocessed as described in Fig. 2, with P = patients.

- 1: $train_set, test_set \leftarrow$ train/test split P (stratified 95%/5% split)
- 2: $folds \leftarrow$ split $train_set$ into 5 stratified, disjoint subsets for cross validation.
- 3: **for** $s \in \{ADC, T1c, T2, FLAIR\}$ **do**
- ▷ Run hyperparameter tuning trials using 5-fold CV for sequence s
- 4: $trials_s \leftarrow$ Hyperband(ResNet50, $folds$)
- ▷ Find trial with lowest mean validation loss over all folds
- 5: $best_trial_s \leftarrow \arg \min \{ \text{mean}\{\text{loss}[t, f] : f \in folds\} : t \in trials_s \}$
- 6: $h_s \leftarrow$ hyperparameters used in $best_trial_s$
- 7: $resnet_s \leftarrow \text{ResNet50}().fit(train_set, h_s)$
- ▷ Verify that $resnet_s$ generalizes well enough to *warrant* explanation.
- 8: $\hat{y}_s \leftarrow resnet_s.predict(test_set)$
- 9: Estimate accuracy, AUC, etc. of $resnet_s$ from $\hat{y}_s$
- 10: **end for**

Algorithm 2: Explain IDH Classifiers Using TCAV Analysis.

Require:

- Concept labels: $\mathcal{C} = \{\text{cobwebbed, freckled, striped, ...}\}$
- Concept training datasets from DTD and MS-COCO: $\{X_C\}_{C \in \mathcal{C}}$
- Bottleneck layers selected for TCAV analysis:
 $\ell_{early} = \text{conv2_block2}, \ell_{intermediate} = \text{conv3_block4}, \ell_{late} = \text{conv5_block1}$

And, for each contrast sequence $s \in \{ADC, T1c, T2, FLAIR\}$,

- $X_s^{IDH} =$ contrast sequence s images, preprocessed as described in Fig. 2
- $X_s^{IDH+} = \{\mathbf{x} \in X_s^{IDH} : \mathbf{x} \text{ is positive/wildtype}\}$
- $X_s^{IDH-} = \{\mathbf{x} \in X_s^{IDH} : \mathbf{x} \text{ is negative/mutated}\}$
- A CNN classifier, $resnet_s$, trained by Algorithm 1 on X_s^{IDH}
- Notation:
 $f_{\ell,s}(\mathbf{x}) =$ activation at layer, ℓ in $resnet_s$ on input $\mathbf{x}$
 $h_{\ell,k,s}(\mathbf{a}_\ell) = resnet_s$ class k logit when activation $\mathbf{a}_\ell$ is applied to layer ℓ
 $v_{C,s} =$ CAV for concept C , layer ℓ of $resnet_s$
 $SC_{C,s,\ell}(\mathbf{x}) = \nabla h_{\ell,k,s}(f_{\ell,s}(\mathbf{x})) \cdot (v_{C,s} / \|v_{C,s}\|)$ (concept sensitivity)

- 1: **for** $C \in \mathcal{C}$ **do**
- 2: **for** $s \in \{ADC, T1c, T2, FLAIR\}$ **do**
- 3: **for** $\ell \in \{\ell_{early}, \ell_{intermediate}, \ell_{late}\}$ **do**
- 4: $lr[C, s, \ell] \leftarrow$ train logistic regression classifier on $f_{\ell,s}(X_C)$
- 5: $v_{C,s} \leftarrow$ parameter vector of $lr[C, s, \ell]$
- 6: $TCAV_{C,wt,s,\ell} = \frac{|\{\mathbf{x} \in X_s^{IDH+} : SC_{C,wt,s,\ell}(\mathbf{x}) > 0\}|}{|X_s^{IDH+}|}$
- $TCAV_{C,mut,s,\ell} = \frac{|\{\mathbf{x} \in X_s^{IDH-} : SC_{C,mut,s,\ell}(\mathbf{x}) > 0\}|}{|X_s^{IDH-}|}$
- 7: **end for**
- 8: **end for**
- 9: **end for**

REFERENCES

- [1] W. L. Bi et al., "Artificial intelligence in cancer imaging: Clinical challenges and applications," *CA Cancer J. Clin.*, vol. 69, no. 2, pp. 127–157, 2019.
- [2] J. W. Jin, M. Fatehi, K. Abhishek, M. Mallya, B. Toyota, and G. Hamarneh, "Artificial intelligence in glioma imaging: Challenges and advances," *J. Neural Eng.*, vol. 17, no. 2, 2020, Art. no. 021002.
- [3] L. W. Barsalou, "Perceptual symbol systems," *Behav. Brain Sci.*, vol. 22, no. 4, pp. 577–609, 1999.
- [4] M. Cimpoi, S. Maji, I. Kokkinos, S. Mohamed, and A. Vedaldi, "Describing textures in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 3606–3613.
- [5] B. Kim, M. Wattenberg, J. Gilmer, C. Cai, J. Wexler, and F. Viegas, "Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV)," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, 2018, vol. 6, pp. 2668–2677.
- [6] S. Nobusawa, T. Watanabe, P. Kleihues, and H. Ohgaki, "IDH1 mutations as molecular signature and predictive factor of secondary glioblastomas," *Clin. Cancer Res.*, vol. 15, no. 19, pp. 6002–6007, 2009.
- [7] P. D. Delgado-López, E. M. Corrales-García, J. Martino, E. Lastra-Aras, and M. T. Dueñas-Polo, "Diffuse low-grade glioma: A review on the new molecular classification, natural history and current management strategies," *Clin. Transl. Oncol.*, vol. 19, no. 8, pp. 931–944, 2017.
- [8] L. H. Gilpin, D. Bau, B. Z. Yuan, A. Bajwa, M. Specter, and L. Kagal, "Explaining explanations: An overview of interpretability of machine learning," in *Proc. IEEE 5th Int. Conf. Data Sci. Adv. Anal. (DSAA)*, 2018, pp. 80–89.
- [9] S. Tonekaboni, S. Joshi, M. D. McCradden, and A. Goldenberg, "What clinicians want: Contextualizing explainable machine learning for clinical end use," in *Proc. Mach. Learn. Res.*, 2019, vol. 106, pp. 359–380.
- [10] A. Ahmad et al., "Predictive and discriminative localization of IDH genotype in high grade gliomas using deep convolutional neural nets," in *Proc. Int. Symp. Biomed. Imag.*, Apr. 2019, pp. 372–375.
- [11] B. H. M. Van der Velden, H. J. Kuijff, K. G. A. Gilhuijs, and M. A. Viergever, "Explainable artificial intelligence (XAI) in deep learning-based medical image analysis," *Med. Image Anal.*, vol. 79, 2022, Art. no. 102470.
- [12] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Int. J. Comput. Vis.*, vol. 128, no. 2, pp. 336–359, 2020.
- [13] Y. Zhang, D. Hong, D. McClement, O. Oladosu, G. Pridham, and G. Slaney, "Grad-CAM helps interpret the deep learning models trained to classify multiple sclerosis types using clinical brain magnetic resonance imaging," *J. Neurosci. Methods*, vol. 353, 2021, Art. no. 109098.
- [14] F. Mercaldo, L. Brunese, F. Martinelli, A. Santone, and M. Cesarelli, "Explainable convolutional neural networks for brain cancer detection and localisation," *Sensors*, vol. 23, no. 17, 2023, Art. no. 7614.
- [15] M. Coupet et al., "A multi-sequences MRI deep framework study applied to glioma classification," *Multimed. Tools Appl.*, vol. 81, no. 10, pp. 13563–13591, 2022.
- [16] F. Šeřčík and W. Benesova, "Improving a neural network model by explanation-guided training for glioma classification based on MRI data," *Int. J. Inf. Technol.*, vol. 15, no. 5, pp. 2593–2601, 2023.
- [17] C. Chen, O. Li, D. Tao, A. Barnett, C. Rudin, and J. K. Su, "This looks like that: Deep learning for interpretable image recognition," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8930–8941.
- [18] D. Wang, Q. Yang, A. Abdul, and B. Y. Lim, "Designing theory-driven user-centric explainable AI," in *Proc. Conf. Human Factors Comput. Syst.*, 2019, pp. 1–15.
- [19] P. Chang et al., "Deep-learning convolutional neural networks accurately classify genetic mutations in gliomas," *Am. J. Neuroradiol.*, vol. 39, no. 7, pp. 1201–1207, 2018.
- [20] O. Davoodi, S. Mohammadzadehsamakosh, and M. Komeili, "On the interpretability of part-prototype based classifiers: A human centric analysis," *Sci. Rep.*, vol. 13, no. 1, 2023, Art. no. 23088.
- [21] E. Calabrese et al., "The university of California San Francisco preoperative diffuse glioma MRI dataset," *Radiol. Artif. Intell.*, vol. 4, no. 6, pp. 2–6, 2022.
- [22] D. Sarwinda, R. H. Paradisa, A. Bustamam, and P. Anggia, "Deep learning in image classification using residual network (ResNet) variants for detection of colorectal cancer," *Procedia Comput. Sci.*, vol. 179, pp. 423–431, 2021.
- [23] M. Gao, D. Qi, H. Mu, and J. Chen, "A transfer residual neural network based on ResNet-50 for detection of steel surface defects," *Appl. Sci.*, vol. 13, no. 9, 2023, Art. no. 5260.
- [24] K. Chang et al., "Residual convolutional neural network for the determination of IDH status in low and high-grade gliomas from MR imaging," *Clin. Cancer Res.*, vol. 24, no. 5, pp. 1073–1081, 2018.
- [25] S. Chen et al., "Predicting MGMT promoter methylation in diffuse gliomas using deep learning with radiomics," *J. Clin. Med.*, vol. 11, no. 12, 2022, Art. no. 3445.
- [26] Y. S. Choi et al., "Fully automated hybrid approach to predict the IDH mutation status of gliomas via deep learning and radiomics," *Neuro Oncol.*, vol. 23, no. 2, pp. 304–313, 2021.
- [27] P. Korfiatis, T. L. Kline, D. H. Lachance, I. F. Parney, J. C. Buckner, and B. J. Erickson, "Residual deep convolutional neural network predicts MGMT methylation status," *J. Digit. Imag.*, vol. 30, no. 5, 2017, pp. 622–628.
- [28] J. Yan et al., "Quantitative MRI-based radiomics for noninvasively predicting molecular subtypes and survival in glioma patients," *NPJ Precis. Oncol.*, vol. 5, no. 1, 2021, Art. no. 72.
- [29] M. B. Hossain, S. M. H. S. Iqbal, M. M. Islam, M. N. Akhtar, and I. H. Sarker, "Transfer learning with fine-tuned deep CNN ResNet50 model for classifying COVID-19 from chest X-ray images," *Inform. Med. Unlocked*, vol. 30, 2022, Art. no. 100916.
- [30] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar, "Hyperband: A novel bandit-based approach to hyperparameter optimization," *J. Mach. Learn. Res.*, vol. 18, no. 1, pp. 6765–6816, 2017.
- [31] B. Efron and R. J. Tibshirani, "Confidence intervals based on bootstrap percentiles," in *An Introduction to the Bootstrap*, 1st ed. Boca Raton, FL, USA: Chapman & Hall/CRC, 1994, pp. 168–177.
- [32] T. Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Comput. Vis. (ECCV) 13th Eur. Conf.*, vol. 13, 2014, pp. 740–755.
- [33] C. Molnar, *Interpretable machine learning*, 2nd ed., 2024. [Online]. Available: <https://christophm.github.io/interpretable-ml-book>
- [34] M. Jafari and N. Ansari-Pour, "Why, when and how to adjust your P values?" *Cell J.*, vol. 20, no. 4, pp. 604–607, 2019.
- [35] E. Calabrese et al., "Combining radiomics and deep convolutional neural network features from preoperative MRI for predicting clinically relevant genetic biomarkers in glioblastoma," *Neuro-Oncol. Adv.*, vol. 4, no. 1, 2022, Art. no. vda060.
- [36] S. R. van der Voort et al., "Combined molecular subtyping, grading, and segmentation of glioma using multi-task deep learning," *Neuro Oncol.*, vol. 25, no. 2, pp. 279–289, 2023.
- [37] J. Wu, Q. Xu, Y. Shen, W. Chen, K. Xu, and X. R. Qi, "Swin transformer improves the IDH mutation status prediction of gliomas free of MRI-based tumor segmentation," *J. Clin. Med.*, vol. 11, no. 15, 2022, Art. no. 4625.
- [38] L. S. Hu, A. Hawkins-Daarud, L. Wang, J. Li, and K. R. Swanson, "Imaging of intratumoral heterogeneity in high-grade glioma," *Cancer Lett.*, vol. 477, pp. 97–106, 2020.
- [39] A. Bai, C. K. Yeh, P. Ravikumar, N. Y. C. Lin, and C. J. Hsieh, "Concept gradients: Concept-based interpretation without linear assumption," in *Proc. 11th Int. Conf. Learn. Representations*, 2023, arXiv:2208.14966v2.
- [40] L. Pasquini et al., "Deep learning can differentiate IDH-mutant from IDH-wild GBM," *J. Pers. Med.*, vol. 11, no. 4, 2021, Art. no. 290.
- [41] P. W. Koh et al., "Concept bottleneck models," in *Proc. 37th Int. Conf. Mach. Learn.*, 2020, vol. 7, pp. 5338–5348.
- [42] M. N. Zaeem and M. Komeili, "Cause and effect: Concept-based explanation of neural networks," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, 2021, pp. 2730–2736.
- [43] D. Bau, B. Zhou, A. Khosla, A. Oliva, and A. Torralba, "Network dissection: Quantifying interpretability of deep visual representations," in *Proc. 30th IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 6541–6549.
- [44] A. Ghorbani, J. Wexler, J. Y. Zou, and B. Kim, "Towards automatic concept-based explanations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, Art. no. 832, pp. 9277–9286.
- [45] Yeh, C. K. Kim, B. Arik, S. Li, C. L. Pfister, and T. Ravikumar, "On completeness-aware concept-based explanations in deep neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 20554–20565.